

Google Trends Data for Forecasting

Executive summary

Google Trends has matured from a novelty signal into a serious forecasting input, but the evidence from 2015–2026 is domain-specific and highly conditional on data handling. Across the 20 representative studies selected here, the strongest and most consistent gains appear in short-horizon tourism nowcasting, selected infectious-disease forecasting tasks, and some mixed-frequency macroeconomic applications. The weakest or most fragile results appear when authors use a single poorly chosen keyword, ignore Google Trends' sampling and normalization quirks, or evaluate models only in idealized rather than real-time settings. Official Google documentation is crucial here: Trends data are sampled, anonymized, categorized, aggregated, and normalized to a 0–100 scale within the chosen time window and geography, which means careless downloading and stitching can distort the signal. ¹

The application papers reviewed here point to a broad regularity. Google Trends tends to help most when the forecast target is behaviorally proximate to search activity and when the horizon is short: current-quarter GDP before many official releases arrive; unemployment claims or unemployment rates in near real time; tourist arrivals 1 week to 2 months ahead; infectious disease incidence 1–3 weeks ahead; and building/electricity demand when searches proxy occupancy, holidays, or near-term operational behavior. Studies in these settings often beat autoregressive, SARIMA, ARMAX, random-walk, or no-search baselines, sometimes materially so. ²

The best recent methodological papers also explain why the literature is mixed. Eichenauer et al. show that raw daily Trends are frequency-inconsistent and noisy unless carefully stitched; Medeiros and Pires show repeated downloads of the “same” Google Trends series can produce materially different samples; and Djorno, Santillana, and Yang show that preprocessing to address zeros, sampling variability, trend shifts, and noise can flip Google Trends from harmful to helpful in influenza hospitalization forecasts. This methodological lesson dovetails with the application literature: feature engineering often matters at least as much as the downstream model class. ³

The literature should not be read as a universal success story. There are credible counterexamples. Nagao, Takeda, and Tanaka found that Google Trends did not necessarily improve U.S. unemployment nowcasts under realistic specification constraints, and Limnios and You found no improvement in one-step-ahead housing forecasts after augmenting standard linear models with Google Trends. Those null results are substantively useful: they show that Google Trends is a conditional leading indicator, not a free forecasting upgrade. ⁴

Scope and selection method

This report prioritizes English-language primary sources from peer-reviewed journals and high-quality preprints indexed or surfaced through academic databases and publisher pages associated with Google Scholar, PubMed/PMC, SSRN, arXiv, RePEc/IDEAS, Wiley, Elsevier, Springer, MDPI, and journal landing pages.

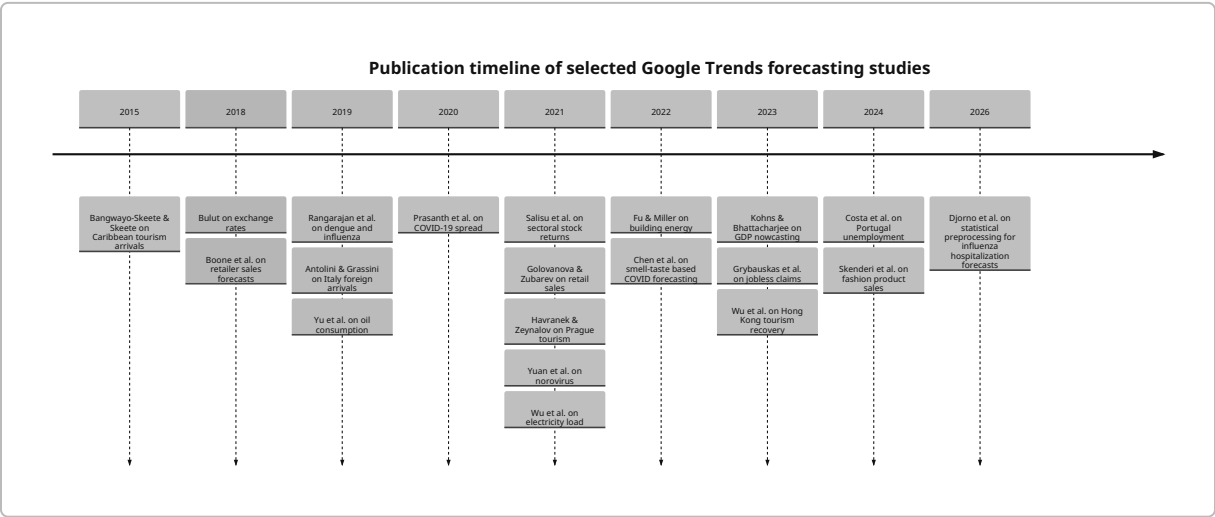
I focused on forecasting or nowcasting studies published from 2015 through June 2026 wherever possible, and selected 20 representative studies spanning economics and finance, epidemiology and public health, marketing and consumer behavior, tourism, and energy demand. I also used official Google Trends help pages and a small number of methodological papers to interpret the application literature correctly. 5

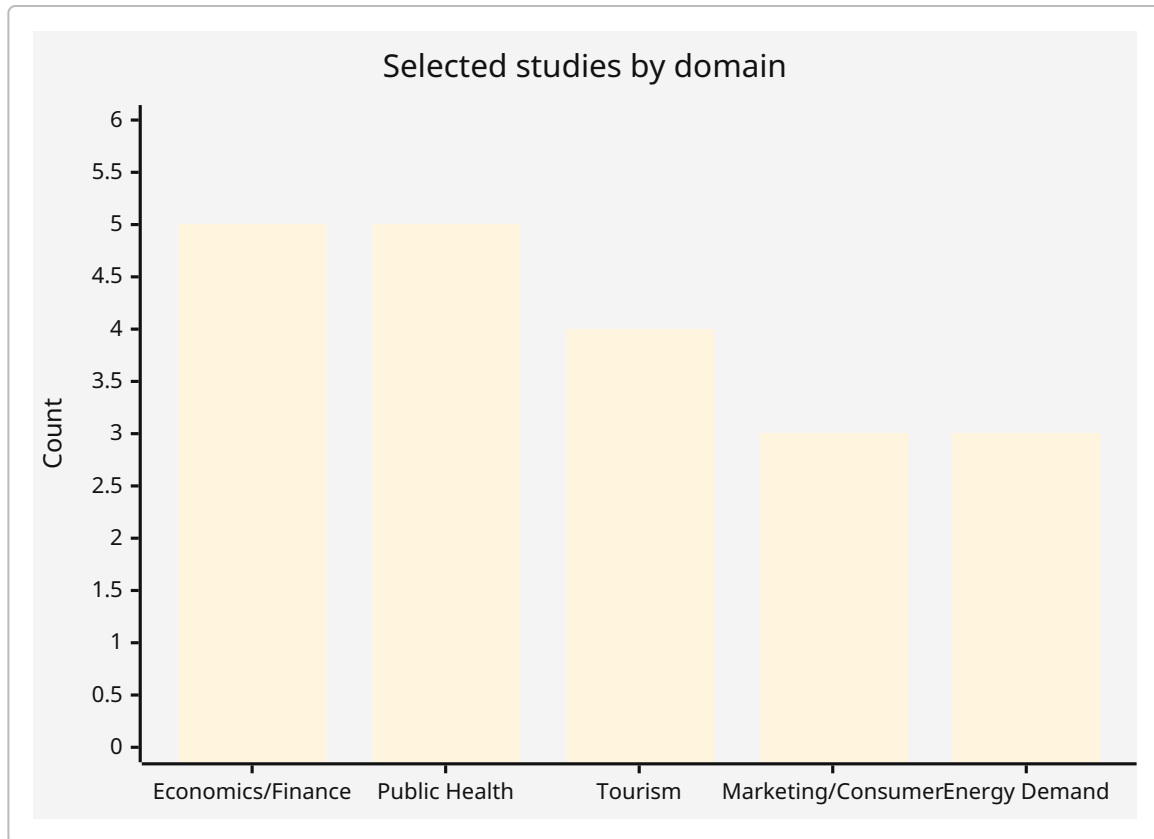
The studies were chosen for a mix of criteria: relevance to forecasting rather than mere correlation analysis, venue quality, methodological clarity, influence or centrality in the literature, recency, and accessibility of enough metadata to evaluate modeling choices and reproducibility. Because accessible citation counts vary across databases and publisher pages, “highly cited” should be read as a prioritization heuristic rather than a strict bibliometric ranking. I therefore favored canonical or frequently referenced papers in strong forecasting outlets, while keeping the sample diverse enough to show both positive and negative findings. 6

Two methodological caveats shape nearly every conclusion below. First, Google Trends normalizes each query by time and location, so values are relative rather than absolute. Second, raw downloads are sampled and may vary across retrievals, while search-term comparisons do not automatically merge misspellings, plurals, synonyms, or multilingual variants unless the researcher deliberately uses more robust topic/ category constructions. These caveats strongly affect term selection, preprocessing, and out-of-sample validation design. 7

Publication landscape and comparison

The selected sample is balanced toward economics and finance and public health, the two areas where Google Trends forecasting is most mature, with tourism close behind. Energy demand and marketing are smaller but increasingly sophisticated, especially where GT is used as an exogenous proxy for latent behavior such as trip planning, shopping intent, or building occupancy. That distribution is consistent with the official design of Google Trends as a fast, category-aware, geographically filterable search-interest signal rather than a replacement for transactional or administrative data. 8





The table below compares the 20 selected studies across the attributes most useful for a forecasting researcher: target, horizon, GT design, model class, benchmark, and whether GT materially helped.

Study	Domain	Sample period	Target and horizon	GT variables	Model class
Kohns & Bhattacharjee 2023, DOI 10.1016/j.ijforecast.2022.05.002 ⁹	Economics/finance	real-time U.S. GDP nowcasting sample; exact span not stated in abstract	Real U.S. GDP growth, early-quarter nowcasts	Large-dimensional GT term set; mixed frequency	BSTS with horseshoe and other shrinkage priors
Grybauskas et al. 2023, DOI 10.3390/economies11050130 ¹⁰	Economics/finance	Full, pre-pandemic, pandemic splits; 222 GT terms	U.S. initial jobless claims; near-term nowcasting	222 keywords across six dimensions	LSTM plus keyword optimization and dimension reduction

Study	Domain	Sample period	Target and horizon	GT variables	Model class
Costa, Silva & Galvão 2024, DOI 10.1016/j.seps.2024.101963 ¹¹	Economics/finance	2019–2021	Portugal monthly unemployment rates; real-time nowcasting	Daily GT keywords tailored to Portuguese usage	MIDAS regressions
Bulut 2018, DOI 10.1002/for.2500 ¹²	Economics/finance	Jan 2004–Jun 2014	Monthly nominal exchange-rate returns across 11 OECD countries	GT search-query proxies for FX fundamentals	Predictive regressions against structural FX models
Salisu, Ogbonna & Adediran 2021, DOI 10.1002/for.2722 ¹³	Economics/finance	11 major U.S. sectors; exact period not stated in abstract snippet	Sectoral stock returns; out-of-sample prediction	Sector-specific GT as investor-attention signal; asymmetric GT shocks	Predictive return models, single- and multi-factor
Rangarajan, Mody & Marathe 2019, DOI 10.1371/journal.pcbi.1007518 ¹⁴	Public health	Dengue: five countries/ states; ILI: U.S.; monthly and weekly series	Dengue and ILI incidence; short-horizon forecasting	Top 10 GT queries most correlated with “dengue”; GT + EHR for ILI	Sparse autoregressive likelihood ratio
Prasanth et al. 2020, DOI 10.1016/j.chaos.2020.110336 ¹⁵	Public health	Early COVID-19 pandemic; multiple regions	COVID-19 spread; short-term forecasting	COVID-related GT terms selected by Spearman correlation	Hybrid GWO-optimized LSTM; ARIMA comparison
Yuan et al. 2021, DOI 10.2196/24554 ¹⁶	Public health	U.S., New York, California; period not fully stated in snippet	Norovirus timing and incidence signals; near-term outbreak prediction	Terms such as gastroenteritis, watery diarrhea, stomach bug, contagious, travel, party	Cross-correlation and stepwise multiple regression

Study	Domain	Sample period	Target and horizon	GT variables	Model class	
Chen et al. 2022, DOI 10.3389/fpubh.2022.1025658 ¹⁷	Public health	Jan 2020–Dec 2021	Worldwide, India, U.S. COVID-19 cases; 1–4 week forecasts	“loss of smell” and “loss of taste” weekly GT	Multivariate transfer-function model	
Djorno, Santillana & Yang 2026, DOI 10.1016/j.ijforecast.2026.03.001 ¹⁸	Public health	U.S. influenza hospitalizations; exact span not stated in abstract	Up to 3 weeks ahead	Influenza-related GT terms	Hierarchical clustering + smoothing splines + detrending, then ARIMAX and ML models	
Boone et al. 2018, DOI 10.1111/poms.12839 ¹⁹	Marketing/consumer	Online retailer; exact period not disclosed in abstract	Product sales; out-of-sample sales forecasting	Customer search data from Google Trends	Time-series models augmented with GT	
Golovanova & Zubarev 2021, DOI 10.31477/rjmf.202104.50 ²⁰	Marketing/consumer	Aggregate and category-level retail sales; period not stated in abstract	Nowcasting and recursive multi-month retail sales forecasts	GT categories aligned to product groups	Linear regressions with macro variables plus GT	
Skenderi et al. 2024, DOI 10.1002/for.3104 ²¹	Marketing/consumer	5,577 new fashion products sold 2016–2019	New-item sales forecasting without past sales history	Image-based and exogenous GT series tied to item visual attributes	Neural encoder-decoder / multimodal deep learning	
Bangwayo-Skeete & Skeete 2015, DOI 10.1016/j.tourman.2014.07.014 ²²	Tourism	Jan 2004–Dec 2012	Monthly overnight tourist arrivals; up to 12 months ahead	Composite weekly searches for “hotels and flights” from U.S., UK, Canada to 5 Caribbean destinations	AR-MIDAS	

Study	Domain	Sample period	Target and horizon	GT variables	Model class
Antolini & Grassini 2019, DOI 10.1007/s11135-018-0748-z ²³	Tourism	Dec 2009–Nov 2016 in preview	Italy foreign arrivals in accommodation; nowcasting	Lagged GT variables	ARIMA and time-series regression with seasonal dummies and AR components
Havranek & Zeynalov 2021, DOI 10.1177/1354816619879584 ²⁴	Tourism	Jan 2010–Dec 2016	Prague tourist arrivals and overnight stays; 1 week and 2 months ahead	Weekly and monthly GT	MIDAS and models with monthly GT
Wu et al. 2023, DOI 10.1016/j.tourman.2023.104759 ²⁵	Tourism	COVID/post-COVID recovery sample; mainland China to Hong Kong	Tourist-arrival recovery forecasting across multiple horizons	Search query data + online news + economy-related variables	MIDAS, GDFM, SARIMA-MIDAS style combinations
Yu et al. 2019, DOI 10.1016/j.ijforecast.2017.11.005 ²⁶	Energy demand	Global oil consumption; exact period not stated in abstract	Oil-consumption trend and level forecasts	Multiple GT signals linked to oil-consumption drivers	Cointegration + Granger screening, then forecasting models
Wu, Dou & Yue 2021, DOI 10.1016/j.ejpsr.2021.107398 ²⁷	Energy demand	Long Island region; daily forecasting setup	Short-term electricity load forecasts	GT/search-engine indices plus weather variables	HP-filter decomposition + QPSO-LSTM
Fu & Miller 2022, DOI 10.1016/j.apenergy.2021.118343 ²⁸	Energy demand	ASHRAE GEPIII building data	Building energy consumption; operational forecasting	GT topics such as education and Microsoft Excel, used as occupancy proxies	Correlation screening + ML prediction models

Annotated review of representative studies

Economics and finance

David Kohns and Arnab Bhattacharjee (2023). *Nowcasting growth using Google Trends data: A Bayesian Structural Time Series model*. International Journal of Forecasting, 39(3), 1384–1412. DOI: 10.1016/j.ijforecast.2022.05.002. PDF/open text: open-access publisher page. This paper asks a fundamental macro question: can a large, mixed-frequency panel of Google Trends series improve real-time U.S. GDP nowcasts before the standard flow of monthly and quarterly indicators is complete? The answer is qualified but positive. The authors show that BSTS models with stronger shrinkage, especially a horseshoe prior, improve materially on both the original BSTS and the SSVS version, with the biggest gains occurring early in the quarter, exactly when forecasters have the least official data. Target and horizon: real U.S. GDP growth, intra-quarter nowcasts. GT design: a large-dimensional term set; the abstract notes that the terms with the highest inclusion probability capture economic anxiety and wealth effects. Preprocessing and features: mixed-frequency alignment and shrinkage of state variances toward zero to reduce overfitting. Models and benchmarks: original BSTS, SSVS-BSTS, horseshoe-BSTS. Evaluation: real-time out-of-sample nowcasting, though the landing-page abstract does not specify the exact loss metric. Limitations: term inventory and reproducibility assets are not obvious on the landing page. Reproducibility: open-access article, but code/data are not prominently linked on the page reviewed. ⁹

Andrius Grybauskas, Vaida Pilinkienė, Mantas Lukauskas, Alina Stundžienė, and Jurgita Bruneckienė (2023). *Nowcasting Unemployment Using Neural Networks and Multi-Dimensional Google Trends Data*. Economies, 11(5), 130. DOI: 10.3390/economies11050130. PDF/open text: open-access MDPI PDF. This is one of the richest keyword-engineering studies in the unemployment literature. Instead of relying on one or two obvious terms such as “jobs” or “unemployment,” the authors compile 222 keywords across six dimensions, including job search, benefits, mental health, violence and abuse, leisure, consumption/lifestyle, and disasters, and feed the resulting features into LSTM-based nowcasts of U.S. initial jobless claims. Target and horizon: near-term changes in U.S. initial claims. GT design: weekly/multi-period GT series across six keyword dimensions. Preprocessing and features: keyword optimization, dimension reduction, min-max scaling to the range –1 to 1, and separate full/pre-pandemic/pandemic analyses. Models and benchmarks: LSTM against narrower keyword choices and conventional unemployment-nowcasting baselines discussed in the paper. Evaluation: the article focuses on forecast-error comparisons; the abstract emphasizes improved errors but instability in the keyword-jobless-claims relationship over time. Limitations: instability over time is the paper’s own main warning. Reproducibility: open-access article with supplementary material, but the data availability statement itself is “not applicable,” so a full replication package is not apparent. ¹⁰

Eduardo André Costa, Maria Eduarda Silva, and Ana Beatriz Galvão (2024). *Real-time nowcasting the monthly unemployment rates with daily Google Trends data*. Socio-Economic Planning Sciences, 95, 101963. DOI: 10.1016/j.seps.2024.101963. PDF/open text: open-access publisher page. This paper tackles a practical problem often ignored by older GT studies: official unemployment data arrive slowly, but Google data arrive daily. Using Portugal as a case study, the authors show that daily Google Trends combined with MIDAS regressions produces more timely and more accurate nowcasts than the more common workflow that first aggregates GT to monthly frequency and then estimates a standard ARMAX-type model. Target and horizon: monthly unemployment rate, nowcast in real time. GT design: daily, culturally dependent Portuguese search terms rather than generic English keywords. Preprocessing and features: mixed-frequency alignment designed for real-time availability. Models and benchmarks: MIDAS with daily GT versus monthly-GT ARMAX-style benchmarks. Evaluation: comparative nowcast accuracy, especially around the COVID-19 shock. Limitations: the landing page does not list the final keyword set or exact loss metric. Reproducibility: open-access article; data are available on request rather than released publicly. ¹¹

Levent Bulut (2018). *Google Trends and the forecasting performance of exchange rate models*. Journal of Forecasting, 37(3), 303–315. DOI: 10.1002/for.2500. PDF: publisher page; SSRN working-paper version also available. Bulut evaluates whether Google Trends proxies for macro fundamentals can improve monthly exchange-rate forecasts across 11 OECD countries. The paper’s main result is nuanced: GT-based predictors do not magically dominate all structure, but they do improve directional forecasting relative to standard structural models and perform at least as well as purchasing-power-parity benchmarks under Clark–West tests, with better results after the Great Recession. Target and horizon: monthly nominal exchange-rate returns. GT design: query-based proxies for macroeconomic fundamentals relevant to exchange-rate models; the exact final query list is not shown in the abstract. Preprocessing and features: out-of-sample predictive regressions over January 2004–June 2014. Models and benchmarks: GT models versus random walk, PPP, and monetary models. Evaluation: directional accuracy and Clark–West equal-predictive-accuracy testing. Limitations: gains are stronger for some currency pairs than others, and the abstract does not expose the full query set. Reproducibility: public code/data are not evident from the landing pages reviewed. ¹²

Afees A. Salisu, Ahamuefula E. Ogbonna, and Idris Adediran (2021). *Stock-induced Google trends and the predictability of sectoral stock returns*. Journal of Forecasting, 40(2), 327–345. DOI: 10.1002/for.2722. PDF: publisher page; abstract also mirrored on ResearchGate. This paper treats Google Trends as an investor-attention measure and asks whether it can predict stock returns across 11 major U.S. sectors. The authors find three broad results: GT is negatively correlated with subsequent sectoral returns, GT-augmented predictive models beat a random-walk baseline, and the gains become stronger when asymmetry in GT shocks and macro controls are modeled explicitly. Target and horizon: sectoral stock returns, out of sample. GT design: sector-linked investor-attention series, with positive versus negative news asymmetry. Preprocessing and features: return transformations and asymmetric decomposition of the GT signal. Models and benchmarks: single-factor and multi-factor predictive regressions against a random-walk baseline. Evaluation: forecast-evaluation framework on sector-by-sector return prediction. Limitations: the abstract does not expose the exact sample span or keyword construction. Reproducibility: no public code/data link was evident on the pages reviewed. ¹³

Public health and epidemiology

Prashant Rangarajan, Sandeep K. Mody, and Madhav Marathe (2019). *Forecasting dengue and influenza incidences using a sparse representation of Google trends, electronic health records, and time series data*. PLOS Computational Biology, 15(11), e1007518. DOI: 10.1371/journal.pcbi.1007518. PDF/open text: free PMC/PLOS article. This is one of the most methodologically mature disease-forecasting papers in the GT literature because it goes past simple correlation and tackles sparse multivariate forecasting directly. The authors combine lagged incidence data, Google Trends, and electronic health records using an autoregressive likelihood-ratio approach, evaluate dengue forecasts in Brazil, Mexico, Singapore, Taiwan, and Thailand, and evaluate influenza-like-illness forecasts in the U.S. Target and horizon: short-horizon dengue and ILI forecasts. GT design: for dengue, the top 10 queries most correlated with “dengue” in each geography; for ILI, GT variables plus EHR signals. Preprocessing and features: sparse variable selection rather than dense AR or lasso-only shrinkage. Models and benchmarks: ARLR versus ARGO, glmnet/lasso, Kalman filtering, ensemble, and naïve methods. Evaluation: forecast-error measures plus peak timing and peak value. Result: about an 18% forecast-error reduction over ARGO on average. Reproducibility is unusually strong: the paper links replication data in Harvard Dataverse and provides supplementary algorithm material. ¹⁴

Sikakollu Prasanth, Uttam Singh, Arun Kumar, and Vinay Anand Tikkiwal (2020). *Forecasting spread of COVID-19 using google trends: A hybrid GWO-deep learning approach*. *Chaos, Solitons & Fractals*, 142, 110336. DOI: 10.1016/j.chaos.2020.110336. PDF/open text: free PMC article. This paper is a representative early-pandemic study that pairs search behavior with a high-capacity nonlinear model. It uses ECDC case data and COVID-related Google Trends terms, selects useful queries via Spearman correlation, and feeds them into an LSTM whose hyperparameters are tuned by the Grey Wolf Optimizer. Target and horizon: short-term COVID spread across multiple regions. GT design: symptom- and pandemic-related search terms filtered for usefulness. Preprocessing and features: correlation-based term screening before the LSTM stage. Models and benchmarks: hybrid GWO-LSTM compared with ARIMA and simpler deep-learning variants. Evaluation: comparative forecasting performance across regions. Key strength: it shows how GT can be fused with nonlinear sequence models in crisis conditions. Limitation: many early-COVID GT relationships were regime-dependent and may not generalize once media saturation changes search behavior. Reproducibility: article is openly accessible, but I did not find a public code repository on the landing pages reviewed. 15

Kai Yuan, Guangrui Huang, Lepeng Wang, Ting Wang, Wenbin Liu, Haixu Jiang, and Albert C. Yang (2021). *Predicting Norovirus in the United States Using Google Trends: Infodemiology Study*. *Journal of Medical Internet Research*, 23(9), e24554. DOI: 10.2196/24554. PDF/open text: free PMC article. This JMIR paper is valuable because it looks not only at the target term “norovirus” but also at symptom, setting, and behavior terms that might lead or lag actual outbreaks. The authors show that queries such as gastroenteritis, watery diarrhea, and stomach bug move with norovirus searches, while terms such as contagious, travel, and party can appear earlier than case-linked GT signals in some settings. Target and horizon: near-term outbreak indication rather than long-horizon forecasting. GT design: multi-term query set evaluated for lead-lag structure across the U.S., New York, and California. Preprocessing and features: cross-correlation analysis and stepwise multiple regression. Models and benchmarks: regression-based signal extraction compared against actual case series timings. Results: several terms provide usable leading information, but the timing relationship varies by geography and term. Reproducibility: free PMC article; code/data release is not obvious from the abstract page. 16

Jingguo Chen, Hao Mi, Jinyu Fu, Haitian Zheng, Hongyue Zhao, Rui Yuan, Hanwei Guo, Kang Zhu, Ya Zhang, Hui Lyu, Yitong Zhang, Ningning She, and Xiaoyong Ren (2022). *Construction and validation of a COVID-19 pandemic trend forecast model based on Google Trends data for smell and taste loss*. *Frontiers in Public Health*, 10, 1025658. DOI: 10.3389/fpubh.2022.1025658. PDF/open text: free PMC article. This study is one of the clearest examples of symptom-linked GT outperforming generic pandemic keywords. Using weekly worldwide, U.S., and India data from January 2020 through December 2021, the authors model new confirmed COVID-19 cases with Google Trends for “loss of smell” and “loss of taste,” and show reliable 1–3 week-ahead forecasting, with some ability to capture turning points over 4-week windows. Target and horizon: new confirmed case counts, especially 1–3 weeks ahead. GT design: two symptom-focused terms with strong biological credibility. Preprocessing and features: multivariate transfer-function time-series analysis. Models and benchmarks: transfer-function model against observed case series; the paper reports absolute forecast errors rather than only correlations. Results: absolute errors were small on the normalized scale for global and Indian forecasts; U.S. forecasts were noisier but still informative. Limitation: symptom-search behavior weakened over time in many countries as COVID became familiar. Reproducibility: free PMC article, but code is not prominently linked. 17

Candice Djorno, Mauricio Santillana, and Shihao Yang (2026). *Restoring the forecasting power of Google Trends with statistical preprocessing*. *International Journal of Forecasting*, available online 30 March

2026. DOI: 10.1016/j.ijforecast.2026.03.001. PDF/open text: publisher page; article in press. This is the single most important methods-oriented epidemiology paper in the recent GT forecasting literature. Rather than assuming raw GT is ready for modeling, the authors explicitly target the platform's current failure modes—missing values, sampling variability, noise, trend drift, and more zeros—and propose a modular preprocessing stack built from hierarchical clustering, smoothing splines, and detrending. Target and horizon: U.S. influenza hospitalizations up to three weeks ahead. GT design: influenza-related keywords. Preprocessing and features: the preprocessing itself is the main contribution. Models and benchmarks: several statistical and ML models with raw GT, preprocessed GT, and no GT. Results: raw GT often degrades statistical forecasts, while preprocessed GT substantially improves them. Reproducibility: unusually strong; the ScienceDirect page notes that the numerical results were reproduced by the journal's Editor-in-Chief.

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Marketing and consumer behavior

Tonya Boone, Ram Ganeshan, Robert L. Hicks, and Nada R. Sanders (2018). *Can Google Trends Improve Your Sales Forecast? Production and Operations Management*, 27(10), 1770–1774. DOI: 10.1111/poms.12839. PDF: publisher page. This short but influential operations-management note makes a practical argument rather than a grand methodological one: customer search activity can improve sales forecasting for an online retailer. The paper shows that augmenting time-series models with Google Trends search data reduces out-of-sample forecast errors, reinforcing the broader idea that user-generated digital traces can carry actionable demand information before purchases post to transaction systems. Target and horizon: product sales forecasts for an online retailer; exact horizon and sample period are not specified in the abstract. GT design: customer search data from Google Trends, likely product-related terms. Preprocessing and models: time-series augmentation rather than complex ML. Benchmarks: retailer time-series models without GT. Limitation: because the note is short and the retailer is not fully described in the abstract, external validity is hard to judge. Reproducibility: no public code/data were evident. 19

Elizaveta Golovanova and Andrey Zubarev (2021). *Forecasting Aggregate Retail Sales with Google Trends. Russian Journal of Money and Finance*, 80(4), 50–73. DOI: 10.31477/rjmf.202104.50. PDF/open text: article PDF and appendices available from the journal/repository pages. This paper generalizes beyond a single retailer to aggregate retail activity and categories of food and non-food products. It pairs macroeconomic variables with GT categories corresponding to product groups and evaluates both pseudo-out-of-sample nowcasting and recursive multi-month forecasts. Target and horizon: aggregate and category-level retail sales, nowcasts plus several-month-ahead forecasts. GT design: category-level GT features mapped to product groups rather than ad hoc keywords. Preprocessing and features: macro-only model selection first, then GT augmentation. Models and benchmarks: linear regressions against the best macro-only specifications. Results: GT significantly improves food and non-food retail forecasts. Limitation: the abstract does not state the precise geography in the search snippet; one should inspect the full PDF for implementation details. Reproducibility: PDF is available, but a code repository is not obvious. 20

Geri Skenderi, Christian Joppi, Matteo Denitto, and Marco Cristani (2024). *Well googled is half done: Multimodal forecasting of new fashion product sales with image-based google trends. Journal of Forecasting*, 43(6), 1982–1997. DOI: 10.1002/for.3104. PDF: publisher page; arXiv preprint and code/data also available. This study is especially interesting because it addresses a hard retail problem where classical time-series methods are weak: new product forecasting with no sales history. The authors create exogenous GT series from visual and textual cues attached to brand-new fashion items, then combine those GT encodings with product images and metadata in a neural encoder–decoder, showing that the

multimodal GT-augmented model beats strong baselines on a real dataset of 5,577 products sold between 2016 and 2019. Target and horizon: launch-period sales for new fashion items. GT design: image-based and visually translated trend queries associated with product appearance. Preprocessing and features: multimodal encoding of exogenous GT, images, and metadata. Models and benchmarks: deep multimodal model versus state-of-the-art alternatives. Results: best performance overall, with roughly 1.5% WAPE improvement from GT. Reproducibility is excellent by GT-forecasting standards: the VISUELLE dataset and code were released publicly. ²¹

Tourism

Prosper F. Bangwayo-Skeete and Ryan W. Skeete (2015). *Can Google data improve the forecasting performance of tourist arrivals? Mixed-data sampling approach*. *Tourism Management*, 46, 454–464. DOI: 10.1016/j.tourman.2014.07.014. PDF: publisher page or author-linked versions. This is still one of the canonical tourism papers. It builds a composite GT indicator from weekly searches for “hotels and flights” from the three main source countries—United States, United Kingdom, and Canada—to five Caribbean destinations and shows that AR-MIDAS models outperform AR and SARIMA in most 12-month out-of-sample experiments. Target and horizon: monthly overnight tourist arrivals, with forecasts up to 12 months ahead. GT design: weekly composite searches for hotels, flights, and destination-country interest. Preprocessing and features: mixed-frequency aggregation via MIDAS. Models and benchmarks: AR-MIDAS against AR and SARIMA. Results: GT materially improves tourism forecasting, especially when past arrivals alone are weak because demand is unstable. Limitation: destination-specific tuning likely matters. Reproducibility: public code/data were not obvious from the landing pages reviewed. ²²

Fabrizio Antolini and Laura Grassini (2019). *Foreign arrivals nowcasting in Italy with Google Trends data. Quality & Quantity*, 53(5), 2385–2401. DOI: 10.1007/s11135-018-0748-z. PDF: author-uploaded PDF available. Antolini and Grassini focus on a classic official-statistics problem: how to nowcast monthly foreign arrivals at accommodation facilities before the national statistical office releases them. They test whether lagged GT variables can improve both ARIMA and time-series regression models with seasonal dummies and autoregressive components, and the abstract/preview strongly suggests that GT adds useful incremental information. Target and horizon: monthly foreign arrivals in Italy, nowcasting. GT design: lagged GT variables tied to tourism demand. Preprocessing and features: seasonal adjustment through dummies and autoregressive error structure. Models and benchmarks: ARIMA and time-series regression with and without GT. Evaluation: nowcast-error comparisons on the preview pages. Limitation: the preview does not expose the final query list or exact loss metrics. Reproducibility: PDF is available via an author-uploaded copy, but code is not obvious. ²³

Tomas Havranek and Ayaz Zeynalov (2021). *Forecasting tourist arrivals: Google Trends meets mixed-frequency data*. *Tourism Economics*, 27(1), 129–148. DOI: 10.1177/1354816619879584. PDF: publisher page with supplementary appendix. This Prague study is valuable because it directly compares weekly versus monthly use of the same GT information. The authors show that Google Trends improves prediction of monthly tourist arrivals and overnight stays, and that the weekly GT series used through MIDAS outperforms both monthly GT models and models without GT, with useful signal as early as two months or as late as one week before arrivals. Target and horizon: monthly arrivals and overnight stays, 1 week to 2 months ahead. GT design: weekly and monthly GT for tourism-related searches. Preprocessing and features: mixed-frequency MIDAS. Models and benchmarks: weekly-GT MIDAS, monthly-GT models, and no-GT alternatives. Limitation: terms are not listed in the abstract page. Reproducibility: the supplementary appendix is available, but a replication package is not obvious. ²⁴

Jing Wu, Mingchen Li, Erlong Zhao, Shaolong Sun, and Shouyang Wang (2023). *Can multi-source heterogeneous data improve the forecasting performance of tourist arrivals amid COVID-19? Mixed-data sampling approach.* *Tourism Management*, 98, 104759. DOI: 10.1016/j.tourman.2023.104759. PDF/open text: free PMC version. This paper moves the tourism literature beyond “GT alone” and puts search queries into a broader digital-signals framework. Forecasting tourist arrivals from mainland China to Hong Kong, the authors combine search query data, online news data, and economy-related variables, then use MIDAS-style models to exploit multiple frequencies. Target and horizon: tourism-demand recovery at multiple horizons in the post-COVID setting. GT design: destination-relevant search query data paired with news and macro data. Preprocessing and features: multi-source heterogeneous data fusion across daily and monthly frequencies. Models and benchmarks: MIDAS/GDFM-style models against benchmark and single-source alternatives; the paper reports MAE, RMSE, MAPE, MASE, and related measures in its notes. Results: multi-source variants materially strengthen forecast accuracy. Reproducibility: free PMC article, but code/data links are not prominent. 25

Energy demand

Lean Yu, Yaqing Zhao, Ling Tang, and Zebin Yang (2019). *Online big data-driven oil consumption forecasting with Google trends.* *International Journal of Forecasting*, 35(1), 213–223. DOI: 10.1016/j.ijforecast.2017.11.005. PDF: publisher page; abstract on ScienceDirect and ResearchGate. This paper is one of the clearest energy-demand applications of Google Trends. It proposes a two-step workflow: first determine whether candidate GT variables are predictively linked to global oil consumption through cointegration and Granger-causality testing, then inject only the effective terms into forecasting models for both the direction and level of oil consumption. Target and horizon: global oil-consumption forecasting. GT design: multiple GT signals intended to reflect oil-demand drivers. Preprocessing and features: causal screening via cointegration and Granger tests before forecasting. Models and benchmarks: GT-augmented forecasting models versus traditional techniques without GT. Results: significant improvements in both directional and level prediction. Limitation: the public abstract does not list the downstream forecast models in detail. Reproducibility: no public code/data repository was evident. 26

Xinyu Wu, Chun-xia Dou, and Dong Yue (2021). *Electricity load forecast considering search engine indices.* *Electric Power Systems Research*, 199, 107398. DOI: 10.1016/j.epsr.2021.107398. PDF: publisher page; abstract mirrored on ResearchGate. This paper extends the GT literature into operational power systems. Using load and weather data for New York State divisions, especially Long Island, the authors show that search-engine indices carry incremental information beyond standard weather and calendar controls; they decompose GT series with the Hodrick–Prescott filter and then forecast load with a quantum-behaved particle-swarm-optimized LSTM. Target and horizon: short-term electricity load. GT design: relevant search-engine indices correlated with load. Preprocessing and features: Spearman correlation, Granger tests, HP decomposition of GT into trend and cycle. Models and benchmarks: QPSO-LSTM with weather and GT versus weather-only inputs and simpler approaches. Results: GT significantly improves short-term forecasting effectiveness. Limitation: keyword details are not visible in the public abstract snippet. Reproducibility: no public code/data link was evident. 27

Chun Fu and Clayton Miller (2022). *Using Google Trends as a proxy for occupant behavior to predict building energy consumption.* *Applied Energy*, 310, 118343. DOI: 10.1016/j.apenergy.2021.118343. PDF: publisher page and author-posted manuscript available. This is a conceptually elegant paper because it treats GT not as “demand” directly, but as an observable proxy for latent occupancy and usage patterns that are otherwise expensive to measure. Using the ASHRAE Great Energy Predictor III competition dataset, the

authors screen GT topics such as “education” and “Microsoft Excel” for correlation with building energy loads and then test whether those topics reduce forecasting error. Target and horizon: operational building energy-consumption prediction. GT design: topic-level search volumes that proxy building use and occupancy. Preprocessing and features: correlation screening, integration with weather/meta/time features. Models and benchmarks: machine-learning energy models with and without GT. Results: RMSLE reductions bring some buildings close to top-five competition performance, with 20–30% reductions during holidays and 2–5% reductions during site-specific schedule periods. Reproducibility: the paper and manuscript are accessible, but code is not prominently linked. ²⁸

Best practices and common pitfalls

The application papers converge on a simple practical rule: start with a behavioral mechanism, not a keyword wishlist. When the search-to-outcome mapping is tight—symptoms to cases, hotels and flights to arrivals, fashion aesthetics to sales, occupancy-related topics to building loads—Google Trends frequently helps. When the mapping is vague or overgeneralized, it often disappoints. This is why symptom-specific COVID terms such as “loss of smell” and “loss of taste” work better than generic “coronavirus” chatter, and why tourism studies built around travel-intent terms or source-destination composites generally perform well. ²⁹

Search-term selection should therefore be hybrid. A good workflow starts with theory-driven seed terms from the demand channel you actually care about, then expands them with data-driven screening inside the training sample only. The literature offers multiple workable variants: correlation-guided symptom selection in COVID forecasting, top-correlated dengue terms in disease prediction, six-dimension keyword construction for unemployment, and category-level GT features matched to product groups in retail. The official Google help pages also matter here: raw search terms do not automatically cover misspellings, synonyms, plurals, or cross-language variants, so topic/category-based constructions and grouped queries are often safer than single bare terms. ³⁰

Researchers should assume raw GT is noisy until proven otherwise. Google states explicitly that Trends is a sample and that values are normalized by location and time. Eichenauer et al. show that raw daily data are frequency-inconsistent and noisy, and Medeiros and Pires show that repeated downloads can produce different samples and therefore different empirical conclusions. Djorno et al. bring these concerns into forecasting directly: raw influenza-related GT can worsen performance, while preprocessing restores predictive value. In practice that means multiple downloads, anchor-based stitching, smoothing or denoising, and explicit treatment of zeros should no longer be optional in serious forecasting work. ¹

Validation design is another dividing line between robust and fragile papers. The strongest studies use rolling or recursive out-of-sample evaluation against meaningful baselines such as AR, SARIMA, ARMAX, random walk, ARGO, or no-search variants, often under real-time data availability. The weaker strand of the literature over-relies on in-sample correlations or on ex post keyword selection that leaks future information. This is exactly why the null findings in unemployment and housing forecasting are useful: they remind us that a GT signal must clear a real-time baseline, not just look plausible on a chart. ³¹

Two model-selection lessons also emerge. First, mixed-frequency statistical models remain highly competitive when the target is published monthly or quarterly but searches update daily or weekly; the success of MIDAS and shrinkage-BSTS in tourism and macro nowcasting shows that “more neural nets” is not automatically the answer. Second, ML and hybrid models add value when the exogenous feature space

is broad, nonlinear, or multimodal, as in LSTM-based unemployment nowcasts, GWO-LSTM COVID forecasting, image-plus-GT fashion sales, and occupancy-proxy energy prediction. The most reliable workflow is therefore not “always use ML,” but “match the model class to the information structure and benchmark it hard.” ³²

For a researcher designing a new GT forecasting project, I would recommend the following validation checklist. Use a theory-driven seed lexicon and document how each term maps to the forecast channel; prefer Google “topics” or grouped/category constructs where multilingual or ambiguous queries are an issue; download GT repeatedly and average or stitch the series; screen terms only on the training window; benchmark against parsimonious statistical models first; evaluate with rolling-origin or pseudo-real-time splits; and report not only average loss but also directional accuracy, turning-point performance, and robustness to keyword-set perturbations. The literature reviewed here supports every step in that checklist.

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Research gaps and bottom line

Three research gaps stand out. The first is reproducibility. Compared with modern forecasting norms, too many GT papers still stop at publisher PDFs with no released code, no archived downloads, and no frozen keyword dictionaries. A few excellent exceptions exist—Rangarajan et al. provide replication data; Skenderi et al. release code and a public dataset; Eichenauer et al. release an R-based stitching procedure; Djorno et al. even passed editorial reproduction—but these are still exceptions rather than the rule. ³⁴

The second gap is cross-language and cross-geography term design. Many papers still use small, opaque, language-specific keyword sets. That is unavoidable in some country studies, but it makes transfer hard and can bias results toward researcher intuition. More work is needed on structured topic dictionaries, query ontologies, multilingual term expansion, and stable category-level GT features that travel across markets. The strongest hints of what this could look like are the multidimensional unemployment paper, tourism source-destination composites, and product-group/category retail models. ³⁵

The third gap is real-time archival behavior. Google itself explains how Trends can be exported and cited, but it does not provide a built-in archival guarantee for every query version, and several methodological papers show that query retrieval can shift over time. Future GT forecasting papers should therefore archive raw CSV pulls, download dates, geography settings, term/topic identifiers, and any stitching logic alongside code. Without that, many published GT forecasting gains will remain only partially auditable. ³⁶

The bottom line is favorable but disciplined. Google Trends is most useful as a high-frequency behavioral covariate for short-horizon forecasting, not as a standalone crystal ball. It works best when the search behavior is tightly linked to the target, when the data are carefully preprocessed and validated in real time, and when models are tested against strong baselines rather than celebrated for beating correlation-only foils. Used that way, the literature from 2015–2026 shows that GT can deliver real forecasting gains in macro nowcasting, public health, tourism, retail, and selected energy-demand settings. Used casually, it can just as easily add noise. ³⁷

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